ARDUINO

Designing Physical AI Systems

From Sensors to On-Device Intelligence

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Milan 🇮🇹, October 28th, 2026

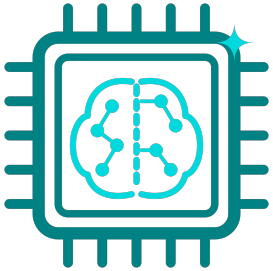
{CODEMOTION} | AI & TECH CONFERENCE

The Evolution of Applied AI

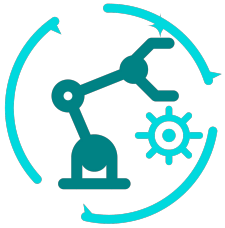
Cloud AI



Edge AI



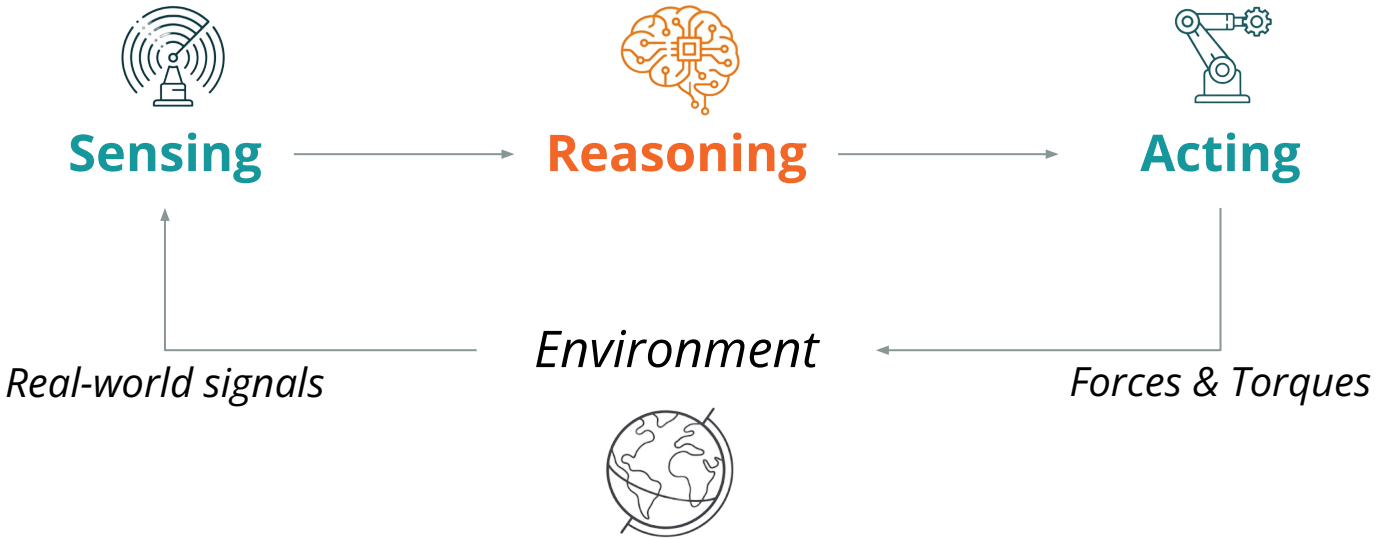
Physical AI



Edge AI is about **local inference**.
Physical AI is about closing the **perception-action** loop.

What is Physical AI?

Physical AI =
Machine Learning + **Embedded Systems** + **Classical Physics**



Physical AI in Action

Logistics





Robotic manipulation

Agriculture





Exoskeletons

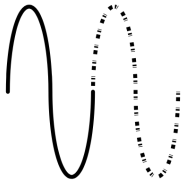
Physical AI is a System Problem

System Success
=
Sensing x **Inference** x Control x Actuation

AI is a component of the system, not the system itself

Physical AI is a System Problem

System Success
=
Sensing x **Inference** x Control x Actuation



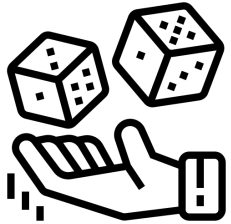
Noisy & incomplete **signals**



Timing constraints



Resource constraints



Probability vs **Safety**

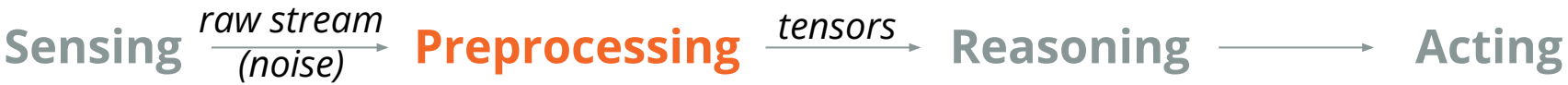
Data is not yet Information

Textbook theory



Impossible in practice!
(Signals ≠ Tensors)

The reality



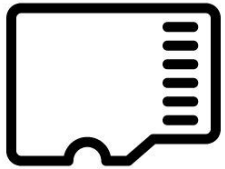
The hidden cost 💰

1. Time delay
2. CPU & Memory bandwidth
3. Information loss

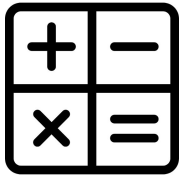
Choosing and Deploying the Model

4 physical constraints

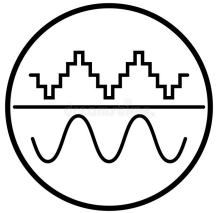
RAM vs Flash



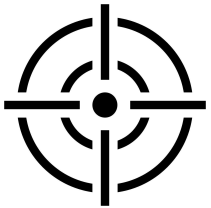
Operator & Kernel Support



Quantization



Accuracy vs Reaction



No Single Architecture Fits Physical AI



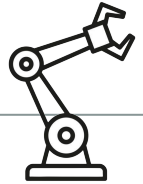
NPU
Inference

- **Energy efficiency** (TOPS/W)
- Dedicated tensor array
- Zero CPU load during model execution



CPU
Orchestrate

- **High throughput** (1-2 GHz)
- Rich OS
- Network stack
- Peripherals management

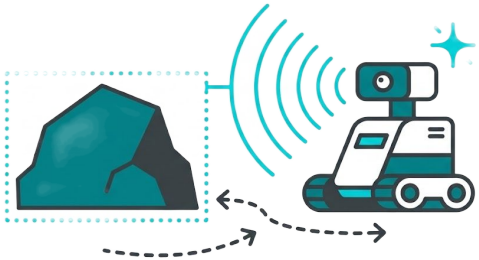


Microcontroller
Act

- **Hard determinism** (µ)
- Zero OS jitter
- Real-time actuation

Latency Is an End-to-End Property

Why a **10ms** inference does not mean a **100 Hz** physical response



$$T_{total} = \underbrace{T_{sensor}}_{40ms} + \underbrace{T_{preprocess}}_{15ms} + \underbrace{T_{inference}}_{10ms} + \underbrace{T_{postprocess}}_{5ms} + \underbrace{T_{actuation}}_{30ms} = 100ms$$

From AI Predictions to Reliable Behavior

Textbook theory



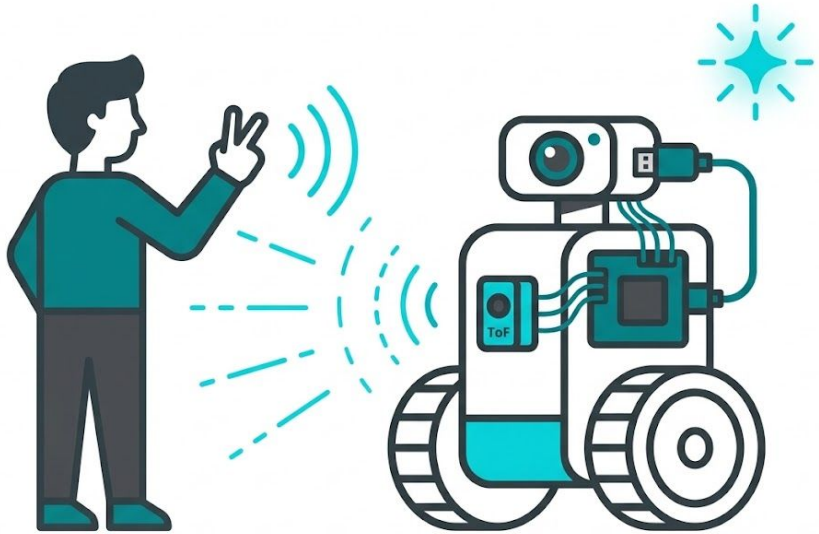
Impossible in practice!
(Probabilities ≠ Physics)

The reality



Deterministic, logical and mathematical supervisor

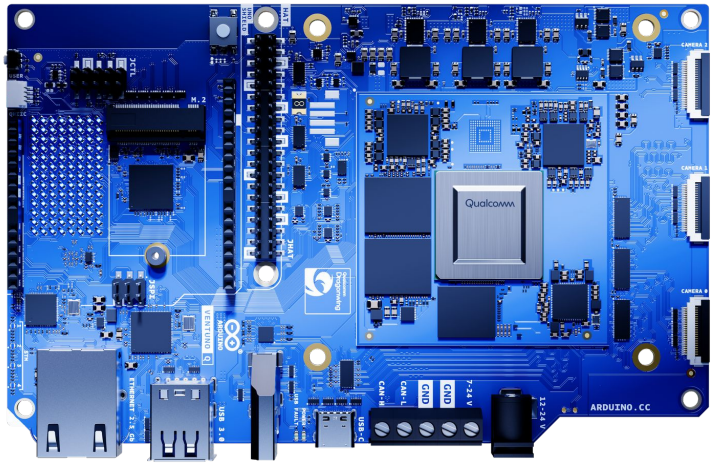
Case Study: Vision-Driven Mobile Robot



Vision-driven control
using hand gestures
✌️🖐️👊

Obstacle detection with
Proximity sensor

The brain: Arduino VENTUNO Q



Powerhouse for **Physical AI** and **Robotics**

Dual brain architecture (CPU&NPU + MCU)

AI ENGINE

40

40 dense Tops

QUALCOMM® HEXAGON™

MEMORY

16

16 GB LPDDR5 RAM

SHARED BY NPU, CPU, AND GPU

CPU

x8

Octa-core

QUALCOMM® KRYO™ UP TO 2.36 GHZ

REAL-TIME

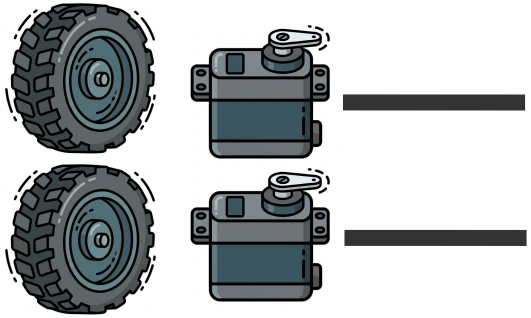
<1

<1 ms

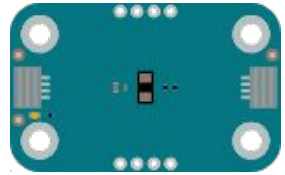
STM32H5F5 AT 250 MHZ

The system

Arduino VENTUNO Q



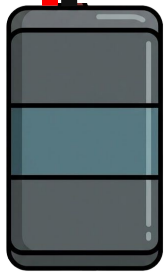
2 x Servo motor + wheels



Modulino
Distance
(ToF Sensor)

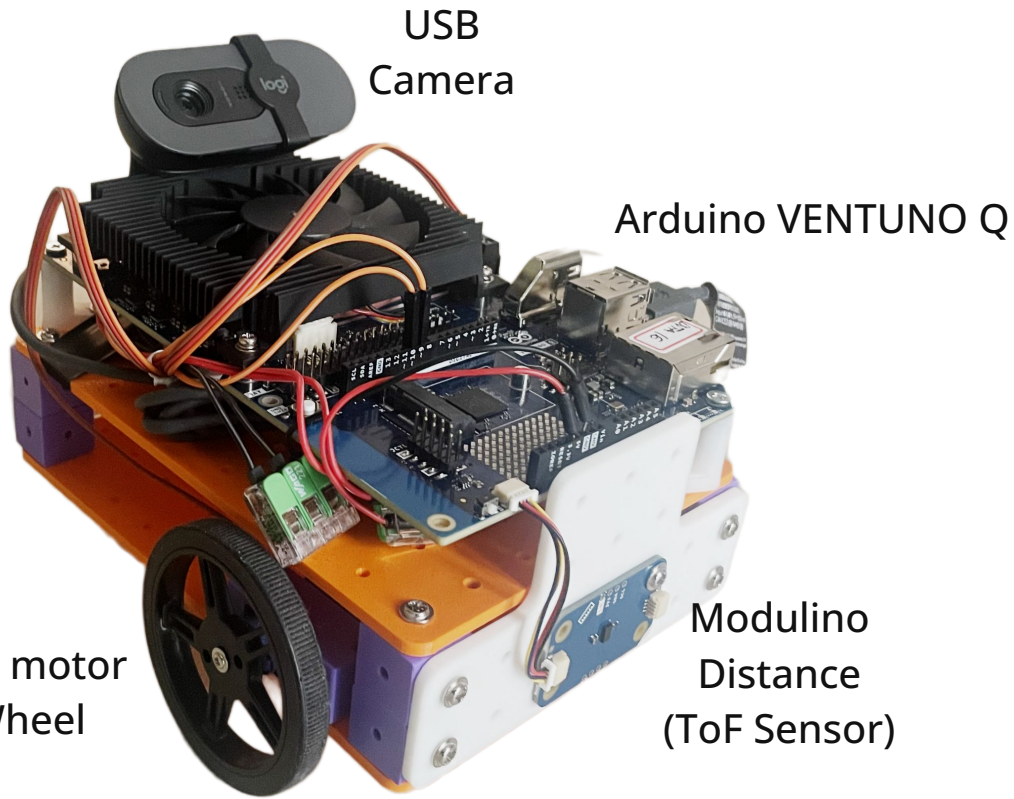


USB
Camera



Lithium
Battery

Vision-Driven Mobile Robot



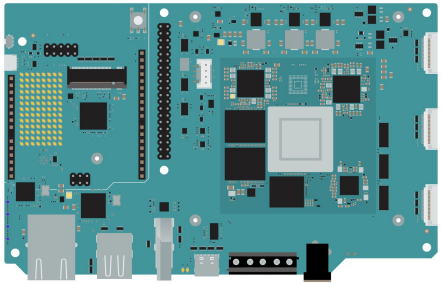
Live Demo: Perception into Action



Image acquisition



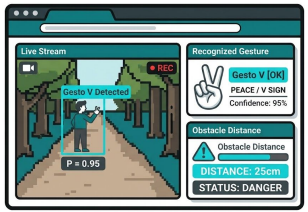
Image classification + Actuation



Motor control

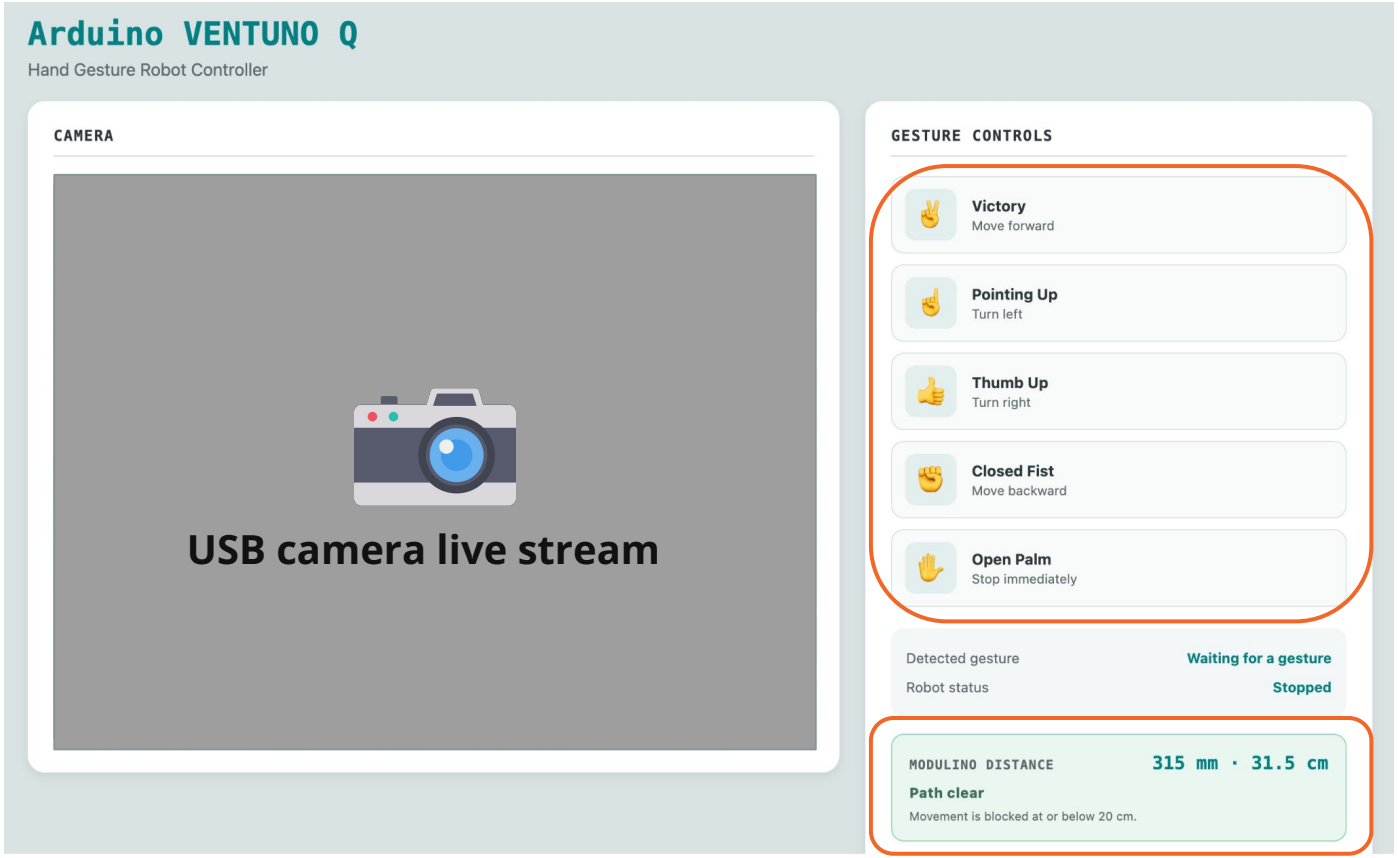


Live dashboard



Camera *streaming*
Gesture *Recognition status*
Obstacle *Detection status*

Live Dashboard



Gesture recognition & action

Obstacle status

Thank you! Grazie!



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Milan 🇮🇹, October 28th, 2026